

Padmanabhan Vijendran

Big Data Developer / Data Engineer

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Profile

- IT professional with **14 years** of experience in developing applications using **Java, Python** and **Big-Data** technologies
- Extensive expertise in working with **Hadoop, Spark 3.x, Spark Structured Streaming, Kafka, Hive** and **HBase**
- **Google Cloud Certified - Professional Data Engineer**
- Strong knowledge of **Python** programming and **Machine learning** concepts
- Proficient in **UNIX commands** and **shell scripting**
- Excellent analytical, communication and monitoring skills
- Adaptable to different technologies and eager to learn new concepts quickly

Professional Experience

Lead Data Engineer, *Royal Bank of Canada*

06/2023 – Present

Project Description: *EODc Regulatory and Compliance Automation*

Toronto, Canada

This initiative focuses on designing and developing a comprehensive framework to automate the generation of critical regulatory and End-of-Day Contract (EODc) reports for RBC's US subsidiaries, including Georgia Bank and City National Bank. As part of this effort, I also spearheaded the creation of the EODc Compliance Checker, a key internal tool that validates daily contract reports to ensure they meet both industry and internal standards. The project leverages a modern technology stack to build robust, scalable, and secure applications that support key business operations and regulatory compliance.

Roles and Responsibilities:

- **Engineered** a scalable, end-to-end data solution using **Java and Apache Spark** to automate the generation of critical regulatory and EODc reports.
- **Architected and developed** the **EODc Compliance Checker**, a Java-based application that validates complex daily contract reports and **ensured compliance with internal and industry standards**.
- **Modernized** application components by migrating from **Java 8 to Java 17** and **Spring 4 to Spring 6**, improving performance and ensuring compatibility with modern standards.
- **Led** the migration of **CI/CD pipelines from Jenkins to GitHub Actions**, **streamlining deployment workflows** and **reducing deployment time**.
- **Managed** and **resolved** high-priority **P1/P2 security vulnerabilities** across multiple applications, maintaining compliance with organizational and industry standards.
- **Optimized** data processing for voluminous **EODc data packages, exceeding 200GB**, using multi-threading techniques to ensure high performance.
- **Maintained** a high level of code quality by implementing **unit tests with over 90% coverage**, integrating **Sonar Lint** for static analysis, and developing **functional test cases** with the **Cucumber Framework**.

- **Collaborated** with **Agile** teams and **business analysts** to translate complex business requirements into technical solutions and ensure the timely delivery of reports.

Senior BigData Developer, *Cognizant Technology Solutions - Client: Ford*

Project Details: Ford engineers use reports generated by an **ETL** framework to design future cars with low manufacturing costs. We are migrating the **ETL** framework from **Sqoop** and **Hive** to **Apache Spark Structured Streaming** so that engineers can access near **real-time** reports.

08/2021 – 06/2023
Mississauga, Canada

Roles and Responsibilities:

- Migrated **ETL** framework from **Sqoop/Hive** to **Apache Spark Structured Streaming** for Ford Engineering.
- Developed **Spark Structured Streaming** application using **pyspark** that **reduced data processing time** and increased **report accuracy**.
- Implemented **ETL** pipelines using **Apache Spark**, **Kafka** and **MongoDB** that enabled engineers to utilize near real-time reports for designing future cars at low manufacturing cost.
- Designed and developed **Java** application to efficiently fetch data from LDAP servers.
- Refactored **Java** applications to improve readability and performance
- Explored new technologies using **POCs** and documented the approaches to implement new project requirements

Senior BigData Developer, *Kelly Services - Client: Credit Agricole CIB*

Project Details: Cash Management Transactions (CMT)

03/2018 – 07/2021
Singapore

Implemented a high-performance, real-time internal payment engine to orchestrate incoming and outgoing payment transactions. Engineered a resilient, event-driven architecture using **Java 8**, **Apache Spark Structured Streaming**, **Kafka**, and **HBase**. Delivered the project following **Agile methodologies** in a dynamic, cross-functional team.

Roles and Responsibilities:

- **Engineered a scalable payment engine** using **Java 8** and **Spark Structured Streaming** to process over **1 million transactions per day** with **sub-second latency**, directly supporting critical business operations.
- **Designed and built a robust data pipeline** with **Kafka** and **HBase** that enabled real-time ingestion, storage, and analysis of transaction data, ensuring data integrity and rapid access for downstream systems.
- **Solved a critical integration challenge** by developing **custom source and sink connectors for IBM MQ** using Spark Structured Streaming. This initiative **improved data reliability and boosted throughput by over 40%** by eliminating a middleware component.
- **Reduced manual effort by 50%** by creating **Python scripts** to automate demo tool execution and performance report generation, allowing the team to focus on core development.
- **Streamlined the CI/CD pipeline** by developing **Jenkins pipelines** with **Groovy** and **shell scripting**, automating the build, test, and deployment of code changes to production.
- **Enhanced code quality and team efficiency** by automating functional tests with the **Cucumber** framework and setting up application monitoring dashboards using the **ELK stack**, providing real-time visibility into performance and system health.

- **Developed a versatile data layer** that enabled seamless communication between the payment engine and various internal applications by translating proprietary formats into a standardized message structure.

Big Data Developer, *Cognizant Technology Solutions - Client: Ford*

02/2016 – 11/2017

Project Details: I worked on the project that generated reports to help Ford Engineer to design a future cars at low manufacturing cost. We implemented **ETL** using **BigData** tools such as **Apache Spark 2.3**, **Hive**, **Sqoop** and **Oozie**.

Chennai, India

Roles and Responsibilities:

- Extracted data from various Oracle databases to **HDFS** and **Hive** using **Sqoop** scripts
- Transformed data to the required schema using **Hive** scripts
- Developed **Java** application to generate Excel reports from **Hive** table data using **Apache POI**
- Explored ETL with **Alteryx**, **Apache Spark** using **POCs**
- Automated manual tasks using **Python** and **shell scripts**

IT Analyst, *Tata Consultancy Services - Client: Target, USA*

01/2011 – 01/2016

Project Details: I worked on the project that processed product images and details for Target website using **ETL Java** application and **SQL** scripts.

Chennai, India

Roles and Responsibilities:

- Developed **ETL Java** application to process product information using **Map-Reduce**
- Wrote **SQL** scripts to transform product related data
- Automated manual tasks using **Python**
- Investigated and analyzed application **support incidents** through to root cause and fix
- Proactively monitored services and responded to alerts raised by customers and **monitoring** systems, ensuring the availability and performance of the **Teamsite** Content Management Systems

Technical Skills

Programming Languages

Java 17, Python 3

BigData Frameworks

Apache Spark, Kafka, Hive, Hadoop

Machine Learning

Algorithms: XGBoost, RandomForest, Logistic Regression, SVM

ML Python libraries: pandas, numpy, scikit-learn, langchain

Automation Scripts

Shell, Batch, Powershell, Selenium

Testing Frameworks

Junit-5, Mockito, Cucumber

Operating Systems

Microsoft Windows 11, Linux, MacOS

JAVA EE Frameworks

Spring Framework 6 (Core, Dependency Injection, Spring Data JPA, Spring Security), Jakarta EE (CDI, Faces, Weld), PrimeFaces (UI development)

Databases

MySQL, PostgreSQL, MongoDB, HBase

Cloud

AWS, Azure, GCP, Oracle Cloud

DevOps

Docker, Jenkins, Github Actions, UrbanCode Deploy, Artifactory

Other Tools

IntelliJ, Eclipse, Git, Gitlab, Putty, ServiceNow, Rally.

AI Tools

Github Copilot, Gemini Code Assist, Claude code

Certificates

- Google Cloud Certified - Professional Data Engineer [↗](#)
- Microsoft Certified: Azure AI Fundamentals [↗](#)
- AWS Cloud Practitioner [↗](#)

Personal Projects

AI Chat Exporter Chrome Extension [↗](#)

Developed a browser extension to export full chat conversations from Gemini and ChatGPT into a clean, portable Markdown file, developed through self-guided learning and rapid prototyping.

Technologies: JavaScript, HTML, CSS, Chrome Extension API.

Interactive ChatGPT Command Line Interface (CLI) [↗](#)

Created a CLI tool for an interactive conversational experience with the OpenAI API, featuring real-time response streaming.

Technologies: Python 3, OpenAI API, command-line interface design.

Machine Learning Odyssey [↗](#)

A portfolio of hands-on Machine Learning projects, including a breast cancer classification model and a house sales price prediction model, showcasing proficiency in data analysis and model development.

Technologies: Python, Scikit-learn, Pandas, NumPy, XGBoost, Jupyter Notebook.

Courses

GCP Data Engineering, *Google Qwik-Labs* [↗](#)

Education

Bachelor of Engineering - Electronics and Communication, *Anna University*

2006 – 2010
Chennai, India